

Aviral Agarwal

aviralagarwal0306@gmail.com | (347) 878-9012 | [YouTube: AviralRobotics](#) | [Instagram: @aviralrobotics](#)

Education

East Brunswick High School, East Brunswick, NJ

Rising junior with straight A's throughout academic career.

Profile

- Aspiring biomedical engineer passionate about combining technology with medicine
 - Strong background in programming, mechanical engineering, and electronics projects
-

Achievements

- Consistent honor roll and academic excellence throughout high school
 - Varsity athlete in Cross Country and Track and Field
 - Self-taught in programming, electronics, mechanical engineering, and game development.
-

Technical Skills

- CAD and Mechanical Design: (Onshape and Autodesk Inventor)
- 3d Printing
- Programming: Java, C#, Lua, Python, TI-Basic
- Game Development: Godot, Unity, Roblox Studio
- Media: Video editing, Custom asset creation (sprites, animations)
- Electronics: Arduino (sensors, motors)

Projects

YouTube Channel: Aviral Robotics

- Created and maintained a STEM-focused YouTube channel documenting original robotics and engineering projects.
- Link: [AviralRobotics](#)

Arduino Projects

- **Computer Vision Robotic Hand:**
Developing a 3D printed robotic hand with individually controlled fingers using computer vision
- **Crane Game:**
Designed, lasercut, and assembled a crane game that used an Arduino to control 2 servos, an electromagnet, and LEDs.
- **Obstacle Avoiding Car:**
Designed and 3d printed a car that uses an ultrasonic sensor and two motors to drive autonomously while detecting and avoiding obstacles
- **Temperature-Controlled Fan:**
Programmed an Arduino to power a DC fan in response to temperature, using sensor integration and motor control.
- **Radar System:**
Built servo driven ultrasonic sensor that triggers LEDs when objects are detected within range.
- **Robotic Arm:**
Designed and assembled a robotic arm using three servos, capable of basic gripping, rotating, and object manipulation.

Ongoing Biomedical Engineering Projects

- **Running Form Analyzer**
 - Uses computer vision to analyze a person's running form, providing data on aspects such as cadence, stride length, and other running mechanics.
- **Running Insole**
 - Uses pressure sensors to collect and analyze foot strike data while running, including force distribution, ground contact time, and asymmetry between legs

Game Development

- **TI-Calculator Games:**
Developed “Pong” and an original object-catching/dodging game in TI-Basic
- **Godot Game:**
Designed and programmed a survival action game featuring a weapon system, AI-controlled enemies, in-game shop, custom graphics, and difficulty progression.
- **Career Island:**
Created a video game for an FBLA (Future Business Leaders of America) event that teaches the player about different work experiences
- **Current Projects:**
 - Sports Game
 - Developing a multi-sport minigame collection in Godot, focusing on varied mechanics and player progression.

Extracurricular Activities

- **Bravura Youth Orchestra:** Cello, honors orchestra, participate in competitions
- **Varsity Cross Country & Track:** Competitive runner and mentor to underclassmen
- **FTC Robotics Team:** Mechanical subteam
- **Teacher at PALS Learning Center:** Spent the summer teaching students about STEM related concepts
- **Clubs:**
 - FBLA (Future Business Leaders of America)
 - State finalist in Computer Game and Simulation Programming
 - Part of the Social Media Committee
 - HOSA (Health Occupations Students of America)
- **African Library Project:** Worked with ALP (African Library Project) to raise over 500 books for a library in Malawi, Africa
- **Soles4Souls Shoe Drive:** Collected over 70 pairs of shoes to donate to people in need